

Cloud-Based IoT MONITORING Of PILLBOX USAGE



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Medication adherence is a term describing the extent to which patients take medications as prescribed by their healthcare providers. Low medication adherence is detrimental to all major healthcare parties—from patients to healthcare providers to insurance companies—eventually adding millions of dollars to the healthcare deficit. Patients who fail to adhere to a strict medication regimen remain ill longer, become frequent users of the emergency room, and often return to the hospital for issues that could have been prevented by maintaining medication protocols.

Medication non-adherence refers to patients who routinely fail to take their medication as prescribed, and is arguably

the largest problem in healthcare. Suboptimal adherence leads to increased costs, lost opportunities and premature deaths.

Wireless and low-power embedded computing technology has enabled the design of a real-time intelligent pillbox to increase medication adherence. Basically, it is a real-world solution using a box with advanced pill detection and wireless connectivity to a cloud infrastructure. The infrastructure aggregates the information and reminds patients to take their medication at appropriate times. It also alerts healthcare providers to any non-adherence.

CloudPill pillbox

Developed by Advel Technologies, Bangalore, the CloudPill electronic pillbox is

basically an Internet of Things (IoT) solution that helps people to monitor their medication on a continuous basis. In healthcare industry, being cloud-enabled offers competitive advantages such as doctor-patient engagement, alert by various methods like SMS and calls, and patients' database. CloudPill device improves on existing systems by providing mobility, frequent and automatic data collection, more detailed information about non-adherence and medication errors, and the familiar interface of a drug-store pillbox. It can be extensively used in elderly care homes, smart homes, hospitals, etc.

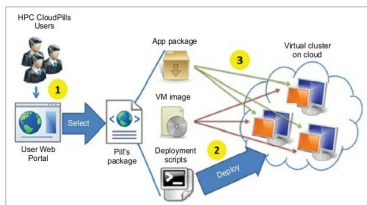


Fig. 1: CloudPill system

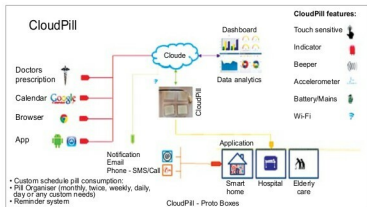


Fig. 2: CloudPill system

An intelligent medication adherence solution using an intelligent pillbox with pill detection capability and wireless connectivity to a cloud infrastructure